Circuit Overview

Components

1. Ultrasonic Sensor
2. Thermistor on ADC/DAC board
3. LED
4. Servo Motor
5. Raspberry Pi Zero

The Raspberry Pi reads the analog temperature data of the room from the thermistor on the ADC/DAC board via I2C, and distance from the ultrasonic sensor. Based on these inputs, it controls the servo motor which simulates a fan and the LED's brightness, which is proportional to the temperature of the room.

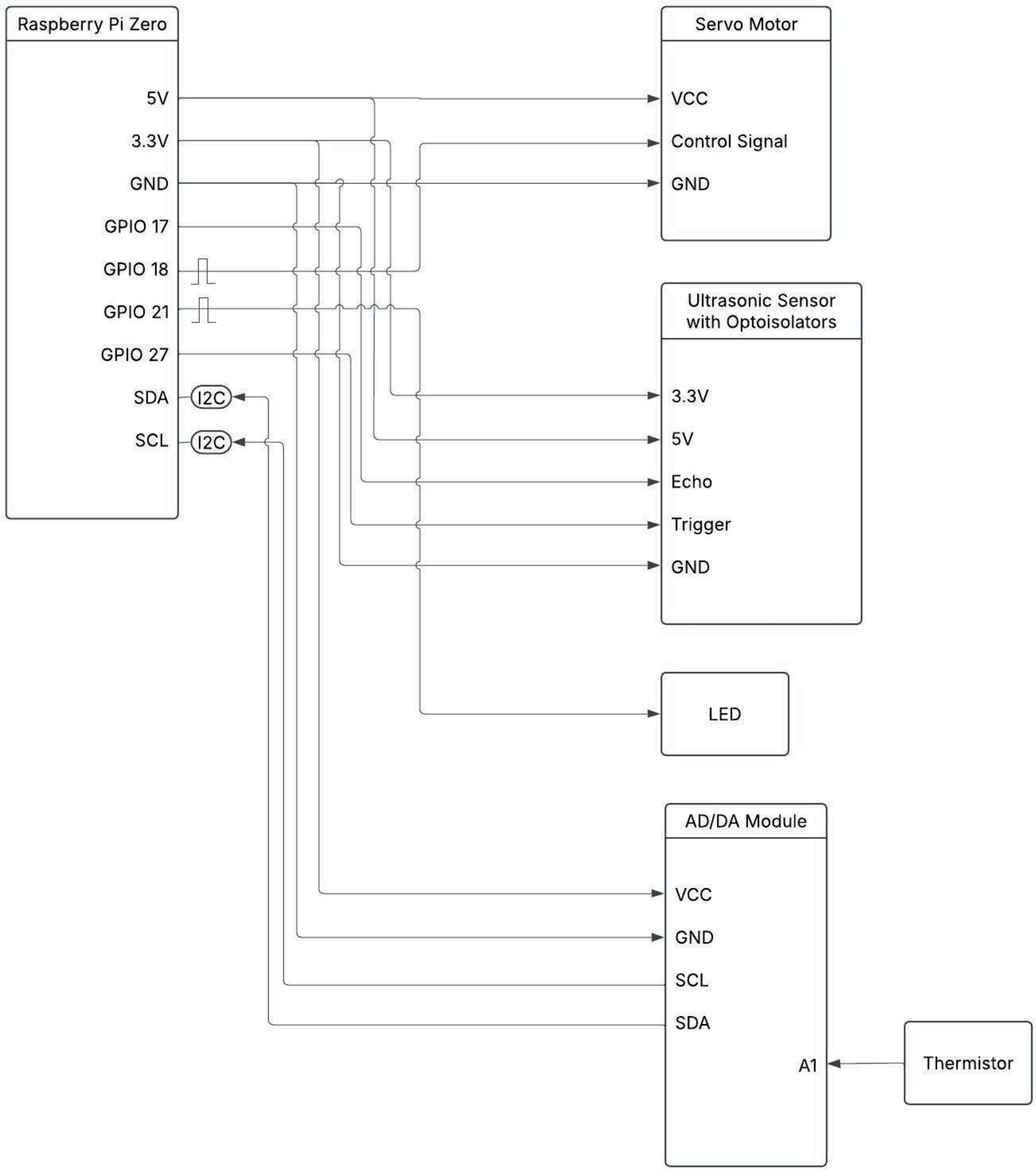
Possible Scenarios

1. Room is cool
   When the room is cool, the servo motor is idle as there is no need for the fan to be on. The LED is either off or very dim.

2. Room is warm
   When the room is warm, the servo motor begins movement from left to right to simulate the fan, and moves at a slow speed. If the ultrasonic sensor detects the presence of a person, the fan speed increases to maximum speed. The LED is at medium brightness.

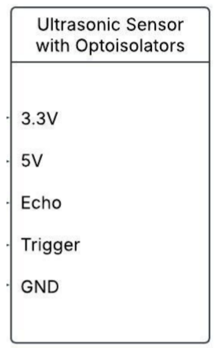
3. Room is hot.
   When the room is hot, the fan speed is at maximum, with or without a person in the room. The LED is bright.

Block Diagram

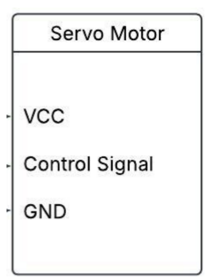


<u>Block Templates</u>

Ultrasonic Sensor Block Template
Device name: Ultrasonic Ranging Module
Model number: HC - SR04
Supply voltage range: 5V
Input type(s): Digital 10uS TTL pulse
Output type(s): Digital TTL pulse proportionate to distance
Interface: Digital
Description: Measures distances ranging from 2cm to 4m.
Symbol:

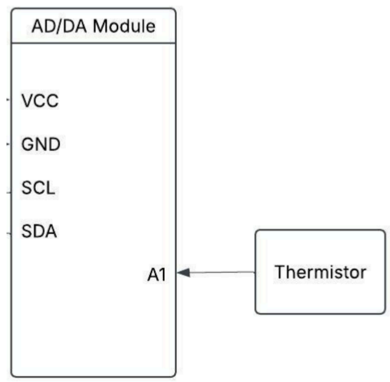


Servo Motor Block Template
Device name: Servo Motor
Model number: SG90
Supply voltage range: 4.8 V (~5V)
Input type(s): PWM, 50Hz, duty cycle <= 20%
Output type(s): N/A
Interface: N/A
Description: Servo motor that can rotate approximately 180 degrees (90 in each direction).
Symbol:

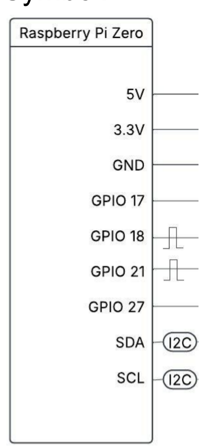


Thermistor via AD/DA Module Block Template
Device name: Thermistor

Model number: PCF8591
Supply voltage range: 3.3 V
Input type(s): Analog
Output type(s): Analog
Interface: I2C
Description: Reads ambient temperature, outputs a voltage inversely proportional to temperature
Symbol:



Raspberry Pi Zero Block Template
Device name: Raspberry Pi Zero
Model number: W/WH
Supply voltage range: 5V
Input type(s): SPI / UART / I2C
Output type(s): PWM / I2C
Interface: I2C
Description: Central conntroller
Symbol:



<u>Block Explanations</u>

Raspberry Pi
Controls the system by reading sensor data, processes logic and controls outputs

Thermistor
Detects temperature changes in the environment (via PCF8591)

PCF8591
Converts analog thermistor signal to digital for Pi to read via I2C

Ultrasonic Sensor
Detects if a person is nearby based on distance measurement

Servo Motor
Acts as the fan, moves slowly or rapidly depending on inputs

LED
Indicates temperature level visually using PWM brightness